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I am currently pursuing BTech 3rd year in Computer Science and also enrolled in the online Bsc in Programming and Data Science program offered by IIT Madras. I'm passionate about software development, with a focus on designing functional web apps using Python, Flask, and up-to-date database systems. Working on this project allowed me to gain practical experience in creating and launching a complete full-stack application with real-world use cases.

Description:

The project is of creating an online Vehicle Parking Management System where users will be able to register, book parking spaces, and see their history, while admins will be able to administer parking lots, users, and see in-depth revenue and utilization summaries using graphical dashboards. I approached it by modularizing the code into models, routes, templates, ensuring both functionality and user friendly design.

AI Usage: I took the help of AI in the topics of flask and jinja which is a backend project that is of approximately 30%.

Technologies used:-

Python, Flask, Jinja2, HTML, CSS, Bootstrap, Plotly, SQLite which is programmatically created Flask SQLAlchemy for database models and ORM.

Reasons of using these technologies:

Python: I utilized python as the primary back-end programming language as it is simple, readable and has an immense library ecosystem

Flask: I utilized flask to manage back-end logic, map routes, handle sessions, and render templates as it is a lightweight Python web framework

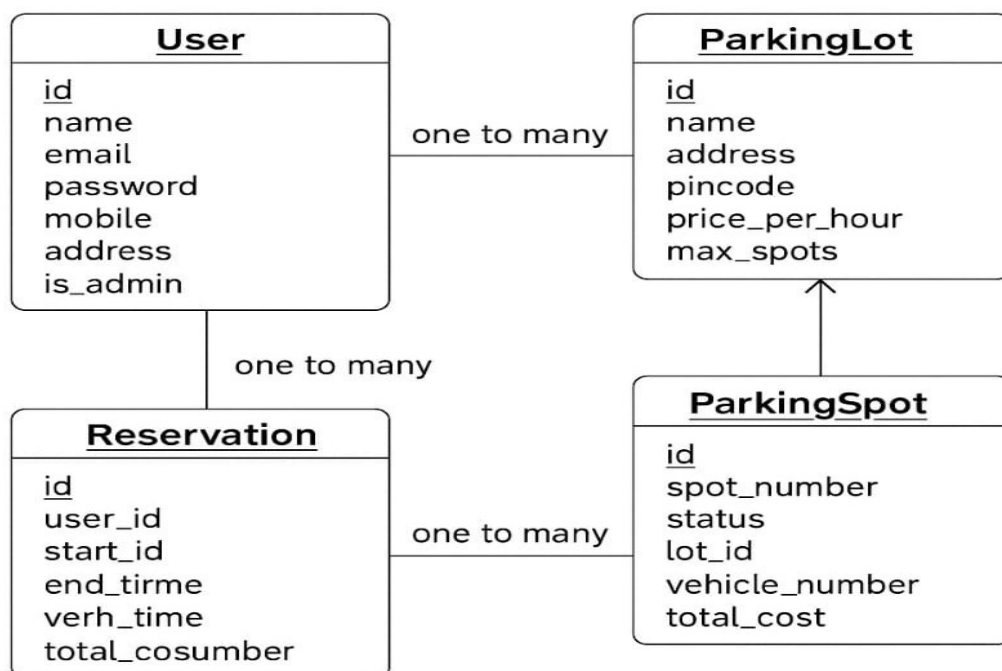
SQLite: SQLite is a light, file-based relational database that is used to store application data like user information, parking lots, spots, and bookings.

Jinja2: Jinja2 which comes from Flask is used to render HTML pages dynamically based on data passed from the back end. It embeds Python logic within HTML in order to iterate over data.

HTML, CSS: HTML is used to organize the web pages, whereas CSS is used to style them, both of them collectively defined the interface's layout, typography, colors, and responsiveness.

Bootstrap: I used Bootstrap as a front-end framework that makes it easier to design responsive and mobile-friendly web pages.

Plotly: I used Plotly to show the summary stats as it is a graph library that provides interactive, good-looking statistical charts and bar plots.

DB Schema Design:

Reasons of designing database this way:

Normalisation: To improve data integrity and prevent redundancy, the schema is normalised.

Scalability: Without changing current structure, additional parking lots, users and reservations can be added.

Flexibility: Start time, end time are used in the schema to track both past and current reservations.

Efficiency: Foreign keys facilitate effective user, lot, and reservation relationships as well as querying.

Role Management: The User table's is_admin column helps manage access between regular users and administrators.

Architecture and Features:

The Flask MVC (Model View Controller) architecture is used in the organisation of the Vehicle Parking App V1. The application logic and routing are managed by the app.py file that serves as the main controller. Models.py uses SQLAlchemy ORM to define all database models. The templates/ directory contains HTML templates, while the static or folder contains custom CSS and Bootstrap styles. Dashboards, forms, and summary views are just a few examples of the user or admin functions that each template relates to.

Features Implemented:

Registering as a user and logging in: Flask sessions are used for a secure session-based login. The administrator's credentials are preset.

Features for the admin: Parking spots and lots can be added, edited, or deleted, see every user details, do a filter search, and remove users and summarize system data with revenue.

Features for Users: Reserve and vacate the spots in the lots, confirm reservation using the car number and also viewing reservations history and summarise usage using graphs.

Additional Enhancements: Editing user and admin profiles, modalities for confirming sensitive actions and also the dashboard user interface using Bootstrap 5.

Video: https://drive.google.com/file/d/1Y275vUNVQkXrjeJsgORu0jcVxV91VD7u/view?usp=drive_link